ECE 302: Probabilistic Methods in ECE
Summer 2023
(Updated 2 June, 2023)

Course Information

Course Title: Probabilistic Methods in Electrical and Computer Engineering
Course Number & Section: ECE 30200-WN1
CRN: 14503
Meeting time: 11:00 am – 12:00 pm, MTWRF (Online, Synchronous) @ https://tinyurl.com/ece302sum23
Credit hours: 3
Course web pages: Brightspace – "Summer 2023 ECE 30200-WN1 LEC"
Gradescope – "Summer 2023 ECE 30200-WN1 LEC"
Piazza – Signup Link: http://piazza.com/purdue/summer2023/ece302

Prerequisites: (Undergraduate level MA 26200 Minimum Grade of D- or Undergraduate level MA 26600 Minimum Grade of D- or Undergraduate level MA 36600 Minimum Grade of D-) and Undergraduate level ECE 30100 Minimum Grade of D- [may be taken concurrently]

Instructor(s) Information & Office Hours

Instructor: Kent Gauen
Office Location: Zoom @ tinyurl.com/gauenk
Email: gauenk@purdue.edu
Online Office Hours: By Appointment

Graduate Teaching Assistant: Biangjian Yang
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Undergraduate Teaching Assistant: Dulani Wijayarathne
Office Location: TBD
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Course Topics Description

An introductory treatment including probability of events, discrete and continuous random variables, multiple random variables, sums of random variables and long-term averages, and elementary random processes. Applications involving uniform, Gaussian, exponential, geometric and related random variables. Introduction to parameter estimation and hypothesis testing. Discussion of wide-sense stationary random processes, including correlation functions. Course examples are drawn from signal processing, wireless communications, system reliability, and data science.

Goals for Learning

Enduring Outcomes (the course outcomes I want to stick with you long after you have finished this course)**:**

- EO-1: Students will understand the purpose, benefits, and limitations of constructing probabilistic models for uncertain events.
- EO-2: Students will understand how the mathematical framework of random variables can be used to model uncertain events.
- EO-3: Students will understand how to analyze the implications of such probabilistic models in order to make decisions which support a desired goal in a given engineering problem.
- EO-4: Students will understand the concept of a random process and how this concept relates to random variables.

Important-to-Know Outcomes (the more specific skills that are important in supporting the enduring outcomes)**:**

- IK-1: Students will be able to understand and apply the mathematical framework for what constitutes a valid probability measure.
- IK-2: Students will be able to thoroughly characterize the probability distribution of a single event using probability distribution functions and related metrics, including moments and variance.
- IK-3: Students will be able to recognize practical situations where basic probability distributions (such as Bernoulli, Binomial, Uniform, or Gaussian distributions) are useful models for an uncertain event.
- IK-4: Students will be able to characterize the probability distribution of multiple events.
- IK-5: Students will be able to characterize the conditional probability distribution of one event given the outcome of another event.
- IK-6: Students will understand what constitutes a random process.
- IK-7: Students will understand how to find mean and autocorrelation functions of a random process.
- IK-8: Students will be able to provide a basic commentary on the reasoning and limitations of how a given probabilistic model is used to make an engineering design decision.

Learning Objectives (the specific skills which will be tested/evaluated throughout the course):

- LO-1: Students will be able to solve elementary word problems involving a single random event/random variable. (EO2, EO3)
- LO-2: Students will be able to solve elementary word problems involving multiple random events/random variables. (EO2, EO3)
- LO-3: Students will be able to create example problems which describe real situations in which probabilistic methods, including at least one of the specific random variable distributions discussed, can be used to derive a solution. (EO1, EO2)
- LO-4: Students will be able to describe the related assumptions and/or potential shortcomings of using a particular probabilistic model in a proposed way. (EO1, EO3)
- LO-5: Given a description of a random process, students will be able to find the mean and autocorrelation functions for that random process. (EO4)

How to Succeed in this Course

- Prepare for class by skimming the textbook sections listed for the class session.
- Come to class expecting to be asked to participate in working on problems and discussing them with peers and/or the instructor.
- Study for exams by reviewing the assigned homework problems and trying additional problems from the textbook, both from in-text examples and the problems at the end of the chapters.
- Pay attention to any emails from the instruction team. This is an online class, so much of the important information will be communicated via email and through Brightspace. I will do my best to keep communications concise and focused on important information.

- Communicate any issues or questions you have early. It is much harder to accommodate last-minute requests or questions.
- Start homework early. The course will move very quickly, so it will be very easy to fall behind.
- Make sure you can dedicate adequate time to the course. This is a full 3-credit hour class, and it is condensed into 8 weeks, so we will move twice as quickly as you would typically experience during a fall/spring semester.

What to Expect from the Instructor

- I will do my best to create a learning environment in which: each student feels welcomed; respect is encouraged between you, all other students, and myself; I challenge you to actively participate during class sessions.
- I will try to respond to email communications within 24 hours (except on Sundays).
- Questions on Piazza will receive feedback within 24 – 48 hours from the teaching staff.
- I will do my best to accommodate unforeseen, difficult, or unique circumstances for students as long as they are communicated as early as reasonably possible.

Required Texts, Software, & Devices

- Required Textbook:
  - Stanley H. Chan, *Introduction to Probability for Data Science*, 1st edition, 2021, e-book. Available at https://engineering.purdue.edu/ChanGroup/eBook.html
- Additional Reading (optional):
  - Alberto Leon-Garcia, *Probability, Statistics, and Random Processes for Electrical Engineering*, Prentice Hall, 3rd edition, 2008.
- Python programming will be used
- Zoom
- Webcam
- The ability to upload scans of your exams

Lectures

Link to Zoom Meeting for Lectures and Instructor Office Hours
https://tinyurl.com/ece302sum23

I will be conducting live lectures over Zoom at the scheduled class time (11:00am – 12:00pm MTWRF; you can use the link above). I strongly encourage you to attend the live lectures, but attendance at the live lectures is not required. The lectures will be recorded and shared via Brightspace (to access recording, go to our Brightspace course page and click "Course Tools" on the black ribbon at the top, and then click "Kaltura Media Gallery").

Grading Policies

Final Grade Calculation

The course will be graded according to the following categories and weights:

- Homework (30%)
- Exam 1 (20%)
- Exam 2 (20%)

- Final Exam (30%)

Final letter grades will be assigned on a curve.

Assignments

Homework

Assignments: There will be eight HW assignments. The points value and due date for each assignment are listed below. In total, there will be 71 HW points which make up 30% of your final grade. While homework is due every Friday at 11:59pm EST, we will accept submissions until Sunday evening.

- HW0 – 1 pt. – Due Wednesday, June 14 at 11:59pm EDT
- HW1 – 10 pts. – June 16
- HW2 – 10 pts. – June 23
- HW3 – 10 pts. – June 30
- HW4 – 10 pts. – July 7
- HW5 – 10 pts. – July 14
- HW6 – 10 pts. – July 21
- HW7 – 10 pts. – July 28

Submitting HW: You may hand-write or type all solutions as best fits the problem, scan/take photos of any hand-written work, then assemble your full work for the assignment into a single PDF. You will upload your work to Gradescope; in the process of uploading, you must assign the appropriate pages of your submission to each question**.** Gradescope provides a simple way for you to match pages of work with each question so that we as graders can easily find your work for each question. Failing to correctly match your pages with each HW question will result in a 5% grade deduction for that assignment. For those unfamiliar with Gradescope, an instructional help video is available here: https://www.gradescope.com/get_started#student-submission.

Cheating Policies: It is acceptable to use resources other than the class materials (book, lectures, notes, etc.) or to collaborate with other students while working on HW assignments. However, you must cite any resources used outside of the course materials. For instance, at the top of your assignment, you could write "worked with Marcus Smith on problems 1-5" or "found solution at https://math.stackexchange.com/questions/4154261/calculate-ex4" (if it is an online resource, you must give the full URL). In addition, you must always write your own solution/program for each problem. If no sources/help is cited, it will be assumed that all work was done with no extra resources beyond the text and lectures. Should we determine that homework answers were copied either from other students, solutions online, or other resources without proper citation, this will be considered cheating – no credit will be given for the assignment and the incident will be reported.

Late HW Policy: Late HW submissions will be accepted for 24 hours past the listed due date and will receive a 5% grade deduction for the assignment. No late submissions will be accepted beyond 24 hours after the listed due date.

HW Solutions: Solutions will be shared in Brightspace by 11:59 am EDT on the Monday following the Saturday when the assignment was due.

Exams

Exam Dates: There will be two midterm exams and one final exam, scheduled on the following dates:

- Midterm Exam 1: Thursday, June 29, 2023; 8:00 – 9:00 pm EST
    - Alternative time (by approval only): Friday, June 30, 2023; 8:00 – 9:00 am EST
- Midterm Exam 2: Thursday, July 20, 2023; 8:00 – 9:00 pm EST
    - Alternative time (by approval only): Friday, July 21, 2023; 8:00 – 9:00 am EST
- Final Exam: TBD (We must wait for the scheduled exam time to be announced by the university)

Resources during exams: Exams will be "open-book". You will be allowed to use class materials (textbook, lectures, notes) and any other *pre-existing* resources (homemade note sheets, pre-existing internet resources, etc.). You are not allowed to collaborate with any other person (student or non-student) in any way during the exam. Just like with homework, you must cite any resources that you used during the exam period on your exam PDF submission. Here are some examples of acceptable and unacceptable resource use on exams.

- Examples of acceptable actions:
    - Looking at the class textbook or notes
    - Making a personal sheet of notes to be used during the exam
    - Looking at a different textbook (with proper citation)
    - Looking at pre-existing websites (with proper citation)
- Examples of unacceptable actions:
    - Working with another student in any way
    - Using an online calculator to compute integrals/derivatives (all work must be shown by hand to convince me that you are able to do the math on your own)
    - Posting exam questions to Piazza
    - Posting exam questions to a Q&A website such as Stack Exchange or Chegg

How the exams will be administered: We will use Zoom to proctor all exams. A few days before each exam, I will share the Zoom link for that exam meeting as an announcement on Brightspace. You will be required to be present in this Zoom meeting during the exam time. You must have your Zoom name as your full name (as it appears in Brightspace) and you must have your camera on and facing you as you work. You must stay on the Zoom meeting for the entire time you are working on the exam and for the entire time you are uploading your work.

Exams will be distributed using Gradescope using two separate assignment links: one for viewing the exam, and one for uploading your work. First, you will view the exam using a Gradescope assignment titled "MT1 Exam View". You will have 60 minutes to view the exam and write down your solutions. Once the 60 minutes have passed, the "MT1 Exam View" assignment will close and you must stop all writing. Then a new assignment called "MT1 Exam Upload" will be made visible, and you will upload your scan to this assignment (just like a homework assignment). You will have 20 minutes after the 60-minute exam period to scan and upload your work. When uploading to Gradescope, you must match your scanned pages to each question in Gradescope – failure to do this will result in a 5% grade deduction on that exam.

Missed Exam Policy: There will be no make-up exams. If you absolutely must miss a midterm exam, you must notify the instructor at least two days before you are scheduled to take the exam. In very special circumstances, and only if approved by the instructor at least two days prior to the scheduled exam, the percentage from the midterm will be shifted to the final exam. We will discuss more details if this becomes necessary.

Early in the course you will be required to commit to the main exam time slot on Thursday at 8:00 pm EST or contact the instructor to request approval to use the alternative time slot on Friday at 8:00 am EST. You may only use the alternative time slot if given prior approval by the instructor early in the course. You will not be allowed to change your selected exam time slot except under extreme circumstances and with prior approval from the instructor.

Communicating with the Instructor(s)

Email: When communicating with me (Kent) over email, please include the course number and the topic in the email subject line (for instance, "ECE 302 HW 1"). I will do my best to respond promptly (usually within 24 hours). I am most available to respond quickly on weekdays and am typically unable to respond to emails on Sundays.

When asking a question, please use the following points to guide you to the best place to ask your question:

- Any questions which could benefit other students should be asked as a public question on Piazza. Personal questions which are only relevant to you should be asked via email.
- Any communications about exam scheduling should be directed to the instructor, Kent Gauen.

Office Hours: Please use the following links to attend office hours at the times describe at the beginning of this document. Office hours will not be recorded.

- Instructor Kent Gauen's office hours link (Zoom): tinyurl.com/gauenk
- TA Poulomi Pradhan's office hours link: TBD

Regrade Requests: Regrade requests for homework or exams must be submitted through Gradescope. You will have one week after an assignment or exam's grade is returned to submit regrade requests.

Incompletes

A grade of incomplete (I) will be given only in unusual circumstances. To receive an "I" grade, a written request must be submitted prior to July 23, and approved by the instructor. The request must describe the circumstances, along with a proposed timeline for completing the course work. Submitting a request does not ensure that an incomplete grade will be granted. If granted, you will be required to fill out and sign an "Incomplete Contract" form that will be turned in with the course grades. Any requests made after the course is completed will not be considered for an incomplete grade.

Academic Integrity

Please read the "Homework" and "Exam" sections under "Grading Policies" for specific details on what is and is not allowed on Homework and Exams. In general, anything that may give a student an unfair advantage over others will be considered dishonest conduct. Examples include (but are not limited to):

- Sharing any information with other students during an exam, including through online communication channels
- Using forbidden material or devices during an exam
- Viewing and/or working on an exam before or after the official time allowed
- Requesting a re-grade of work that has been altered
- Submitting work that is not your own without giving proper credit to its source

All cases of academic dishonesty will be reported to the Office of Student Rights and Responsibilities and will result in punishment. Possible punishments include, but are not limited to, a score of zero on work related to the cheating incident, a failing grade for the course, and in severe cases, expulsion from the university.

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing or by calling 765-494-8778. While information may be submitted anonymously, the more information that is submitted provides the greatest opportunity for the university to investigate the concern. Please follow the Purdue Honor Pledge: "As a boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue."

Students with Disabilities

Any students with a "Letter of Accommodation" from the Disability Resource Center should share this letter with me during the first week of the course or as soon as it is acquired. Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let me know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: **drc@purdue.edu** or by phone: 765-494-1247.

Nondiscrimination Statement

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. Link to Purdue's nondiscrimination policy statement.

Emergency Preparation

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email. You are expected to read your @purdue.edu email on a frequent basis.

Mental Health Statement

- **If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try WellTrack.** Sign in and find information and tools at your fingertips, available to you at any time.
- **If you need support and information about options and resources**, please see the Office of the Dean of Students for drop-in hours (M-F, 8 am- 5 pm).
- **If you're struggling and need mental health services**: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS office of the second floor of the Purdue University Student Health Center (PUSH) during business hours.

Netiquette

Your instructor and fellow students wish to foster a safe online learning environment. All opinions and experiences, no matter how different or controversial they may be perceived, must be respected in the tolerant spirit of academic discourse. You are encouraged to comment, question, or critique an idea, but you are not to attack an individual. Our differences, some of which are outlined in the University's nondiscrimination statement below, will add richness to this learning experience. Please consider that sarcasm and humor can be misconstrued in online interactions and generate unintended disruptions. Working as a community of learners, we can build a polite and respectful course ambience. Please read the Netiquette rules for this course:

- Do not dominate any discussion. Give other students the opportunity to join in the discussion.

- Do not use offensive language. Present ideas appropriately.
- Be cautious in using Internet language. For example, do not capitalize all letters since this suggests shouting.
- Avoid using vernacular and/or slang language. This could possibly lead to misinterpretation.
- Keep an "open-mind" and be willing to express even your minority opinion.
- Think and edit before you push the "Send" button.
- Do not hesitate to ask for feedback.

Violent Behavior Policy

Purdue University is committed to providing a safe and secure campus environment for members of the university community. Purdue strives to create an educational environment for students and a work environment for employees that promote educational and career goals. Violent Behavior impedes such goals. Therefore, Violent Behavior is prohibited in or on any University Facility or while participating in any university activity.

See the University's full violent behavior policy for more detail.

Diversity and Inclusion Statement

In our discussions, structured and unstructured, we will explore a variety of challenging issues, which can help us enhance our understanding of different experiences and perspectives. This can be challenging, but in overcoming these challenges we find the greatest rewards. While we will design guidelines as a group, everyone should remember the following points:

- We are all in the process of learning about others and their experiences. Please speak with me, anonymously if needed, if something has made you uncomfortable.
- Intention and impact are not always aligned, and we should respect the impact something may have on someone even if it was not the speaker's intention.
- We all come to the class with a variety of experiences and a range of expertise, we should respect these in others while critically examining them in ourselves.

Course Evaluation

In the final weeks of the course, you will be provided with an opportunity to evaluate this course and your instructor. Purdue uses an online course evaluation system. You will receive an official email from evaluation administrators with a link to the online evaluation site. You will have up to two weeks to complete this evaluation. Your participation is an integral part of this course, and your feedback is vital to improving education at Purdue University. I strongly urge you to participate in the evaluation system.

Disclaimer

This syllabus is subject to change. Any official changes to the syllabus will be announced via email and/or Brightspace.

Course Schedule

This is a rough outline of how we will progress through the course materials. This schedule will likely be adjusted as we move through the course and better figure out the pace that works for everyone.

Date	Topic	Chapters	Notes	Homework
12-June	Introduction & Geometric Series	1.1		
13-June	Integration, Combinatorics	1.3, 1.5		
14-June	Set Theory & Probability Space	2.1, 2.2		HW 0 Due
15-June	Axioms & Conditional Probability	2.3, 2.4		
16-June	Independence & Bayes Theorm (pt 1/2)	2.5, 2.6		HW 1 Due
19-June	Bayes Theorem (pt 2/2) & Random Variables	2.6, 3.1		
20-June	The PMF	3.2		
21-June	The CDF	3.3		
22-June	Expectation	3.4		
23-June	Moments	3.5		HW 2 Due
26-June	Bernoulli & Geometric	3.6, 3.8		
27-June	Binomial	3.7		
28-June	Poisson	3.9		
29-June	Midterm Review	1 - 3	Midterm 8pm EST	
30-June	PDF	4.1		HW 3 Due
3-July	Expectation	4.2		
4-July	No Class		Independence Holiday Observed	
5-July	CDF	4.3		
6-July	Uniform + Exponential	4.5, 4.6		
7-July	Gaussian	4.7		HW 4 Due
10-July	Function of a Random Variable & Inverse CDF Method	4.8, 4.9		
11-July	Joint PMF, PDF, and CDF	5.1		
12-July	Joint Exectation and Covariance	5.2, 5.3		

13-July	Conditional Distributions & Expectation	5.4, 5.5		
14-July	Sum of Two Variables	5.6, 5.7		HW 5 Due
17-July	Random Vectors	5.8		
18-July	Multidimentional Gaussian	5.9		
19-July	PCA	5.11		
20-July	Midterm Review	4 - 5	Midterm 8pm EST	
21-July	Moment Generating Functions and Characteristic Functions	6.1, 6.2		HW 6 Due
24-July	Jensen's Inquality & Chebyshev Inequality	6.3, 6.4		
25-July	Law of Large Numbers & Central Limit Theorem	6.5, 6.6		
26-July	Maximum Likelihood Estimation	8.1	To be modified	
27-July	Maximum a Posteriori	8.2	To be modified	
28-July	MMSE	8.3	To be modified	HW 7 Due
31-July	Poisson Process	Class Notes	To be modified	
1-Aug	Final Exam Review			
2-Aug	No Class		Final Date TBD	
3-Aug	No Class		Final Date TBD	
4-Aug	No Class		Final Date TBD	

* Schedule and assignments subject to change.